UCC 301.COM 422 L3

Choosing a research topic

Research topic, research title and research problem

Research Topic

A topic is what the essay or research paper is about. It provides a focus for the writing. Of course, the major topic can be broken down into its components or smaller pieces. A research topic is broad as 'undergraduate student learning'. In defining your research area, you will need to think about what is your broad area and what are the boundaries of your research area and what is excluded.

For the topic, you should

1. Select an educational matter that is of personal interest or concern to you.

2. Determine whether the topic is researchable, considering ethics, complexity, and time.

3. Compose a tentative working title for your research

Criteria for choosing topic

- Pick a (manageable) topic that interests you. Try to pick a topic that you find interesting. This will ensure the research process is as engaging and fun as possible
- Be flexible: Understand that choosing a research topic is a cyclical process. No topic should be set in stone. Be prepared to change your topic as you search for information, read about your topic, and learn more.
- Search for information on your topic: Search for information resources, such as scholarly articles and books, in the library catalogue and relevant databases. What do you find? If you find too many information resources, your topic might be too broad. To make your topic more specific and narrow, try to zero in on a single aspect of your topic. Narrow your

topic and search again. If you cannot find enough information resources, your topic might be too specific. To broaden your topic, consider removing some of your criteria from your research topic. For example, instead of investigating the use of Instagram by university students at funerals, you might broaden your topic to the use of web 2.0 tools by university students at funerals. Alternatively, you might need to adjust the keywords you have been using – talk to a librarian for help. Broaden your topic and search again.

- Read about your topic: As you read through your chosen information resources, you will learn more about your topic. In response to what you learn, you may wish to tweak your topic further. Don't be afraid to change your topic to reflect what you read.

Research Title

The title summarizes the main idea or ideas of your study. A good title contains the fewest possible words needed to adequately describe the content and/or purpose of your research paper.

This is a precise statement about the research aims and area. An example would be: 'Effects of research development programmes on undergraduate student learning'. You will then need to generate research questions indicating the specific ways you will contribute to this area.

In addition, a research title should have the following characteristics:

1. Short, descriptive and to the point
2. Contain key variables in the research study (investigative areas)
3. Relate to the topic, niche area and discipline
4. Attract the attention and interest of the reader
5. Make clear sense.

Research question

A research question is a clear, focused, concise, complex and arguable question around which you center your research. You should ask a question about an issue that you are genuinely curious about.

The Importance of a Good Research Question

A good research question assists to:

1. Define the topic – whether your focus is on youth suicide, environmental degradation, secularization, etc.

2. Define the nature of the research Endeavour– whether your aim is to discover, explore, explain, describe, or compare.

3. Define the questions you are interested in– whether you are interested in what, where, how, when, why.

4. Indicate whether you foresee a relationship between concepts you are exploring – whether you are looking for impacts, increases, decreases, relationships, corre-lations, causes etc.

CRITERIA FOR A GOOD RESEARCH PROBLEM

In formulating research problems/questions, it is possible that the problems that may be important to some people may not be so to others. It is also possible that some problems are not easy to understand and investigate than others. A good research problem should have the following qualities

1. It should be researchable or verifiable- it should be one which can be investigated through the collection and analysis of data

2. It must have a theoretical or practical significance. This means that it should contribute to your field of research

3. It should be clear. It should be stated clearly and concisely
4. It should be ethical. i.e it should not involve psychological damage to the people being investigated
5. It relates to an academic discipline
6. It has a basis in the research literature
7. It should have potential significance i.e it should be important to you and of consequence to the others
8. It should be that you can carry out within the available time frame and budget
9. Sufficient data should be available or can be obtained to address it
10. It should be one that you can adequately investigate given your research skills, available time and resources

Examples of research questions

1. To what extent are student centred methods used in universities (What would be the topic and title of this study?)
2. What information management strategies were used to manage legal information in the coronial period
3. Do open source content management software perform better than proprietary content management software

SOURCES OF RESEARCH PROBLEMS

Personal Experience

Everybody has in one way or another been exposed to some interesting or challenging situation in the school or outside. This could be lateness, poor performance, and so on. So you may be

required to answer the question “what is causing lateness on my side” or “what is making me perform poorly in this subject”

Deduction from Theories

Deductive explanation arrives at specific conclusions on the basis of generalizations. There are many theories or general principles whose applicability to specific educational problems is not known until tested. Such theories can be found in politics, personality, learning, economics and others.

Literature Review

Scientists believe that experiments can be repeated at different times and in different situations and still yield similar results. Repeating a study helps to increase generalisability and confidence in the findings. Therefore, by reading previous studies done by other researchers, you may find them interesting to be replicated but in different situations.

Practical Issues

Society is always faced with many burning issues which are not easily explainable eg social media ethics, examination malpractices, universal basic education etc. you may decide to take one of these issues and investigate it with a view to finding a solution to the problem

Deductive Reasoning

This is the conclusion that is made from particular to general eg creating of general rule by seeing similarities among specific situation eg jane is good in language (1st premise). Jane is a girl (2nd premise) therefore; girls are good at language (conclusion)

Inductive reasoning

This is the conclusion that stems from general to specific. It points new relationships as one proceeds from general to specific. Eg Girls are dependent therefore Mary is dependent

Attend professional conferences

Many researchers have great success finding new research projects at national or regional conference problems in their discipline.

Seek the advice of experts. This is a simple yet highly effective strategy for identifying a research problem.

Media – issues which are frequently reported in the media can also form basis of research

VARIABLES

A variable is a type of quantity that may take on more than one value eg level of education is a variable because it can be differentiated by distinct values such as no schooling, some primary education, some secondary education, completed secondary education, and college/ university education. There are many distinct types of variables. Two of the most important are the dependent and independent variables.

Independent variables

This is a variable whose effect you would wish to establish in the study. This is the variable you as the researcher can manipulate in order to ascertain whether or not the result you obtain is due to it. Eg Factors affecting punctuality in schools. The factors constitute the independent variables. They cause effect on punctuality.

There are two types of independent variables: **treatment variable** and **attribute variable**

A researcher can change treatment variable whereas attribute variable cannot be altered

Eg in the above example several factors such as distance, sex, age, rain and security may affect punctuality. Age and security are attribute variables while distance and security are treatment variables.

Dependent Variables

These are variables that the researcher measures in order to establish the change effect created on them eg factors affecting punctuality, punctuality is the dependent variable

NB: A research title should contain the key variables. That is the independent and dependent variables

Extraneous (confounding) variable

This is a variable whose effect is not needed in the study. Eg in the example above it has been established that age, security and distance have an effect on punctuality. However, it is possible that some other factors are responsible but the researcher is not interested in them eg. Class level, family background etc. extraneous variables have three sources: physical environment, attributes of participants (characteristics we are not aware of) and procedural variations ie how we plan the study. Extraneous variables should be controlled so that the research is authentic eg by eliminating it, randomization, holding it constant etc

RESEARCH HYPOTHESIS

In research, the problem is transformed into variables and the variables are transformed into hypothesis. A hypothesis is a tentative prediction of the outcome of results i.e a tentative answer to the problem. It is based on the relationship between variables in the study. Eg if V is attitude and DV is performance, the hypothesis will be there is an effect of attitude on performance

Characteristics of a good hypothesis

a. A good hypothesis must be stated clearly and briefly giving the expected relationships between the variables example U<20.
b. Both the independent and dependent variables are included.
c. Based on a sound rationale derived from theory or previous research or professional experience.
d. Consistent with acceptable traits.
e. Testable.
f. Testable in a reasonable period
g. Consistent with the purpose, statement and objective of the study.
h. Simple and precise.

Functions/purpose of hypothesis in research

1. They provide direction, bridge the gap between the problem and evidence needed for its solution.

2. It ensures collection of the evidence necessary to answer the problem passed in the problem statement.
3. It permits the researcher to understand the problem with greater clarity and use the data collected to find solutions to the problems.
4. It forms the framework for the ultimate conclusion of a study.

Types of hypotheses

1. Null hypothesis (H0)

This is one that is stated in negation. It is a statement that states no relationship or difference that exist between the parameter and the statistic being compared to it. EG there is no difference between boarding and day schools

2. Alternative hypothesis (Research hypothesis) (H1)

This is stated to indicate actual expectation or relationship

There are two types of alternative hypothesis: -

i) Alternative directional hypothesis

It is a statement that states that a relationship or difference exists between the given variables and goes further to specify the nature of the relationship or the difference between the variables. The relationship can be stated using greater than, less than, increased, decreased. Alternative hypothesis is denoted by H1or Ha. For example, H1: U>20 or H1U<20

ii) Alternative non-directional hypothesis

This type of hypothesis states that there is a relationship of difference between the stipulated variables but the researcher doesn't know the nature of such a relationship or difference. This type of hypothesis does indicate a relationship or difference but do not indicate the direction of this relationship. For example, H1U ≠ 20 or H1Um≠ UF

Do I always need a hypothesis?

Not all research questions can be easily 'converted' into a hypothesis, nor should they. The definition of a hypothesis limits its applicability for many types of research, and many research questions.

A hypothesis may not be appropriate if:

- You do not have a hunch or educated guess about a particular situation – For example, you may want to study alcoholism in the South Pacific, but you do not feel you are in a position to hypothesize because you are without an appropriate cultural context for educated guessing.
- You do not have a set of defined variables – Your research may be explorative in a bid to name the contributing factors to a particular situation. In the case of alcoholism in the Pacific Islands, your research aim may be to identify the factors or variables involved.
- Your question centres on phenomenological description – For example, you may be interested in the question, 'What is the experience of drinking like for Pacific Islanders?' A relationship between variables does not really come into play.
- Your question centres on an ethnographic study of a cultural group – For example, you might want to ask, 'What is the cultural response to a defined problem of alcoholism in a South Pacific village?' In this situation, force fitting a hypothesis can limit the potential for rich description.
- Your aim is to engage in, and research, the process of collaborative change – In 'action research', methodology is both collaborative and emergent, making predetermined hypotheses impractical to use.

STATING THE RESEARCH PROBLEM

After identifying a research problem, you must articulate it in such a way that its carefully phrased and represents the single goal of a research effort. A problem statement is the description of an issue currently existing which needs to be addressed. It provides the context for the research study and generates the questions which the research aims to answer. The statement of the problem is the focal point of any research. A good problem statement is just one sentence (with several paragraphs of elaboration). For example, it could be: "The frequency of job layoffs is creating fear, anxiety, and a loss of productivity in middle management workers." While this problem statement is just one sentence, it should be accompanied by a few paragraphs that elaborate on the problem. The paragraphs could cover present persuasive arguments that make the problem important enough to study. They could include the opinions of others (politicians, futurists, other professionals); explanations of how

the problem relates to business, social or political trends via presentation of data that demonstrates the scope and depth of the problem.

A well-articulated statement of the problem establishes the foundation for everything to follow in the proposal and will render less problematic most of the conceptual, theoretical and methodological obstacles typically encountered during the process of proposal development. This means that, in subsequent sections of the proposal, there should be no surprises, such as categories, questions, variables or data sources that come out of nowhere: if it can't be found in the problem section, at least at the implicit level, then it either does not belong in the study or the problem statement needs to be re-written.

WHAT ARE KEY COMPONENTS OF THE STATEMENT PROBLEM?

Problem statements the following elements:

- State the general case / problem
- Describe what others have done
- What's missing / where is the gap in knowledge?
- Describe the proposed solution or research objectives/questions
- Specify one or more specific hypotheses
- Should include specific metrics/measurements
- Discuss how their validation addresses the research questions
- Specific results (or research design, if it is a proposal)

These elements should be brief so that the reader does not get lost. One page is enough for a statement problem.

Criteria for Research Problem Statements:

- The statement of the problem should clearly indicate what is to be investigated.
- The actual statement may be in a declarative or in a question form.
- The statement should indicate the variables of interest and the specific relationship between the variables that are to be studied.

Please note: In some qualitative methodologies, a statement of variables will not be possible and should not be done. However, an explanation of the qualitative methodology and the parameters of the research methodology should be explained.

WHAT IS THE PURPOSE OF PROBLEM STATEMENT?

- Represents the reasons/motivation behind your proposal (based on the specific domain of study).
- It specifies the conditions you want to change or the gaps in existing knowledge you intend to fill (this is the specification of the research problem).
- Should be supported by evidence.
- Specifies your hypothesis that suggests a solution to the problem.
- Shows your familiarity with prior research on the topic and why it needs to be extended.
- Even if the problem is obvious, your reviewers want to know how clearly you can state it.

Sample statement of problem one

TITLE: An exploration of absenteeism from non-mandatory synchronous sessions in the virtual learning environment by undergraduate students at Karatina university

The Topic

Distance education via online platforms is a rapidly growing method of education delivery due to its convenience, wide reach, relatively low cost, and ability to support the achievement of learning objectives. Whether the platform is Blackboard, WebCT, Moodle, Angel, or some other learning management system, online education utilizes a variety of common learning tools including discussion boards, drop boxes, automated testing, and wikis. Chief among these tools are live online sessions.

The Research Problem

Live online sessions may be delivered in virtual classrooms from google meet, zoom, or other software programs. Regardless of the software used, student attendance at live online sessions, especially optional ones, can be unpredictable at best. It is a common complaint among the online faculty at a university in the south that many, oftentimes most, of their students do not attend the live online sessions. This study will address the problem of low student attendance at non mandatory virtual classroom meetings in online college courses.

Background and Justification

Offir, Lev, and Bezalel (2008) found the interaction level in a synchronous class, also known as web conferencing, to be a significant factor in the effectiveness of the class. Other researchers describe "the power of a synchronous online system to empower students in conversation and expression (McBrien, Jones, & Cheng, 2009). However, if students do not attend, then they cannot interact nor express themselves.

Deficiencies in the Evidence

According to Skylar (2009), "research concerning the use of newer multimedia technologies, such as interactive synchronous web conferencing tools, is in its infancy and needs further and continued study" (p. 82). McBrien, Jones, and Cheng (2009) stated that "more studies are needed to explore students' perceptions of the synchronous learning experience." A variety of studies have explored the differences in functionalities of the various platforms (Kenning, 2010; Lavolette, Venable, Gose, & Huang, 2010), but they did not get to the heart of why students do or do not attend.

Audience

This study will benefit college and university administrators who can create or revise policies based upon the results. Administrators may even decide to change virtual classroom providers. Faculty may benefit if results indicate a change is needed in their own practices. Finally, the study will benefit online students whose learning experiences will be improved by the findings.

NB: Your statement of problem should be continuous without the sub topics. I have just included them to help you formulate yours.

Sample statement of problem two

RESEARCH TITLE: Task-Oriented Pattern Discovery for Predictive Web User Modeling

An essential task in building personalized and adaptive systems is the automatic discovery of predictive models for user behavior. Existing approaches, such as clustering, correlation analysis, and association discovery, tend to generate shallow patterns which do not capture the full complexity of users' online behavior. Nor can the generated patterns explain the users' underlying interests which lead to specific types of behavioral patterns. To better capture users' underlying interests or information needs, this research introduces the notion of "task". A task is a set or sequence of actions which are likely to be performed commonly by users in order to meet a specific information need or perform a specific function. These tasks are not directly visible, but can be captured and characterized either by a combination of users' interactions with the site and the site's content and structure.

This study hypothesizes that patterns discovered at the task level can provide a better understanding of users' underlying interests, which in turn, can lead to better predictive models. The study proposes a framework for "Task-Oriented Web User Modeling". The study intends to use probabilistic latent variable modeling to automatically discover and quantify user "tasks" and task-level patterns from users' navigation data, as well as from Web site's content and structure data. Experiments will be performed on real Web usage data and movie rating data to validate that the proposed approach results in more accurate and flexible predictive models.

AIM AND OBJECTIVES

Research aim

The term research aim usually refers to the main goal or overarching purpose of a research project. Sentences stating the aim of a project are usually quite brief and to the point

Research objectives

A research aim will usually be followed by a series of statements describing a project's research objectives. Research objectives indicate in more detail the specific research topics or issues the project plans to investigate, building on the main theme stated in the research aim. Normally at least two or three research objectives will be stated. It is good practice to put these in a numbered list so they can be clearly identified later in a proposal or report. Here is an example of a set of research objectives:

Examples of SMART research objectives

As a rule of dumb, there would be one research aim and several research objectives to facilitate the achievement of this aim.

The following is an example:

Research title: Effects of organizational culture on achieving its aims and objectives: a case study of Virgin Atlantic

Research aim: To assess the effects of Virgin Atlantic organizational culture on achievement of its goals.

Following research objectives would facilitate the achievement of this aim

1. Analyzing the nature of organizational culture at Virgin Atlantic by September 1, 2017

2. Identifying factors impacting Virgin Atlantic organizational culture by September 16, 2017

3. Analyzing impacts of Virgin Atlantic organizational culture on employee performances by September 30, 2017

4. Providing recommendations to Virgin Atlantic strategic level management in terms of increasing the level of effectiveness of organizational culture by October 5, 2017

RESEARCH QUESTIONS

In some situations, rather than stating research objectives, researchers will prefer to use research questions. In the example below, the objectives stated in the previous example are reframed as research questions:

Question 1: Is alcohol consumption associated with increased partner violence?

Question 2: Is labour force status (employment, unemployment, not in the labour force) associated with variations in the incidence of partner violence?

Question 3: Are there differences between couples with an extended history of partner violence and couples with only a brief, recent history of partner violence?

Second example

AIM

The aim of this study is to explore absenteeism from non-mandatory synchronous sessions in the virtual learning environment, by undergraduate students in online courses at a Karatina university.

RESEARCH OBJECTIVES

The objectives of this study will be to:

i. Identify the reasons why students attend non-mandatory synchronous sessions in Karatina University.
ii. To find out why students do not attend non-mandatory synchronous sessions in Karatina University
iii. To establish the actions to be taken by university and faculty to increase student attendance at non-mandatory synchronous online sessions

RESEARCH QUESTIONS

i. What are students' reasons for attending non-mandatory synchronous sessions in Karatina University?
ii. What are students' reasons for not attending non-mandatory synchronous sessions in Karatina University?
iii. What actions could the university or its instructors take that would motivate students to increase their attendance at non-mandatory synchronous online sessions?